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Faculty of Physics, TU Dortmund

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Probing Fast Radio Burst Environments: Cross-Correlating Deep and Shallow FRB Surveys with KiDS DR5

Bachelor Thesis Defense

Will Ruppert

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Research Group Rhode / Elsässer, TU Dortmund University
Research Group Hildebrandt, Ruhr University Bochum

Supervisors: PD Dr. Dominik Elsässer Prof. Dr. Hendrik Hildebrandt

Outline

Introduction

Theoretical Background

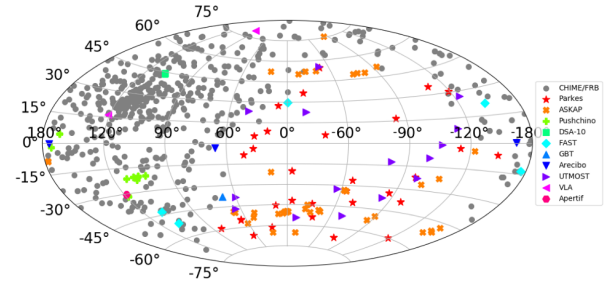
Methodology

Results

Conclusion & Outlook

Fast radio bursts: bright millisecond flashes from far away

- Radio pulses lasting $\lesssim$ ms
- Peak flux densities from **50** mJy to **100** Jy
- Detected from **110** MHz to **8** GHz
- One burst releases as much energy as the Sun in three days
- Received signal $\approx 10^3\times$ weaker than a mobile phone on the Moon
- First detection: Lorimer Burst (2007)
- Second CHIME/FRB catalog: **4539** bursts from **3641** sources, 83 repeaters



Known FRBs in Galactic coordinates (Aitoff projection), up to 2022

Adapted from Petroff et al. 2022; Petroff et al. 2019; Lorimer et al. 2007; CHIME/FRB Collaboration 2026

The dispersion measure reveals an extragalactic origin

Lower frequencies arrive later; the delay defines the dispersion measure (DM)

$$\mathrm{DM} = \frac{2\pi m_e c}{e^2}\,\Delta t \left(\frac{1}{\nu_{\mathrm{low}}^2} - \frac{1}{\nu_{\mathrm{high}}^2}\right)^{-1} \qquad \mathrm{DM} \propto \int_0^L \mathrm{d}s\, n_e(s)$$

Milky Way

Galactic electron-density models:
DM ≲ 50 pc cm^{-3}

Observed FRBs

Often
DM ≳ 1000 pc cm^{-3}

- Excess DM ⇒ sources at cosmological distances
- Approximately isotropic on the sky ⇒ extragalactic population
- FRBs therefore trace the large-scale structure of the Universe

Petroff et al. 2019; Reischke et al. 2021

Two competing progenitor channels

Repeating sources rule out a purely cataclysmic origin for at least part of the population.

Young magnetars (prompt)

- Neutron stars with extreme magnetic fields
- Born in core-collapse supernovae, **3** Myr to **48** Myr after star formation
- Formation rate follows the cosmic star-formation history
- FRB-like burst from Galactic magnetar SGR 1935+2154 (2020)

Neutron star mergers (delayed)

- FRB 20180924A, FRB 20190523A: massive, low-SFR hosts, large offsets
- FRB 20200120E in a globular cluster
- Delay times up to several Gyr
- Population evolves slowly with redshift

This work: the difference is encoded in a redshift-dependent bias $b(z)$

Petroff et al. 2019; CHIME/FRB Collaboration 2020; Wang et al. 2020; Zapartas et al. 2017; Ruiter et al. 2019; Lin et al. 2022

Idea: let the clustering tell us where FRBs live

- Host identification of single FRBs needs arcsecond localizations ⇒ rare
- FRBs trace the large-scale structure ⇒ their angular clustering encodes the bias of the host population
- Host bias depends on the progenitor channel
- Problem: FRB samples are sparse ⇒ strong shot noise
- Solution: cross-correlate with a dense photometric galaxy sample (KiDS-Legacy)

Research question

How well can the FRB host bias be constrained by cross-correlating FRBs with KiDS-Legacy, and can a prompt magnetar origin be distinguished from a delayed neutron-star origin?

Shallow, CHIME-like and deep, SKA-like FRB survey × two progenitor models → Fisher forecast

Rafiei-Ravandi et al. 2021; Reischke et al. 2021

A flat ΛCDM universe sets the distance scale

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right)\right] \qquad 1 + z = \frac{a(t_0)}{a(t_e)}$$

Late-time expansion rate (flat ΛCDM)

$$E(z) \approx \sqrt{\Omega_{\mathrm{m},0}(1+z)^3 + \Omega_{\Lambda,0}}$$

Line-of-sight comoving distance

$$\chi(z) = \chi_{\mathrm{H}} \int_0^z \frac{\mathrm{d}z'}{E(z')}$$

- $\chi(z)$ maps redshift distributions onto the line of sight
- $H(z)/c = \mathrm{d}z/\mathrm{d}\chi$ converts kernels from redshift to comoving distance

Hogg 2000; Planck Collaboration 2018

Tracers are biased versions of the matter field

Density contrast and linear growth

$$\delta(\mathbf{x}) := \frac{\rho(\mathbf{x}) - \bar{\rho}}{\bar{\rho}} \qquad \delta(\mathbf{x}, z) = D_+(z)\,\delta(\mathbf{x}, 0)$$

Power spectrum

$$\left\langle \hat{\delta}(\mathbf{k})\,\hat{\delta}^*(\mathbf{k}') \right\rangle = (2\pi)^3 \delta_D(\mathbf{k} - \mathbf{k}')\,P(k)$$

Linear bias

$$\delta_t(\mathbf{x}, z) = b(z)\,\delta(\mathbf{x}, z)$$

- Tracers form in density peaks
- $b > 1$: more strongly clustered than matter
- $b(z)$ tells us which halos host the FRBs

Bernardeau et al. 2002; Desjacques et al. 2018

Projecting onto the sky: the angular power spectrum

Limber approximation for two tracers A, B (valid for $\ell \gtrsim 10$)

$$C_\ell^{AB} \approx \int \frac{\mathrm{d}\chi}{\chi^2}\, W^A(\chi)\, W^B(\chi)\, P_{AB}\!\left(k = \frac{\ell + 1/2}{\chi},\, z(\chi)\right) \qquad P_{AB}(k, z) = b_A(z)\, b_B(z)\, P(k, z)$$

Radial weight functions

$$W_{\mathrm{FRB}}(z) = b(z)\, n(z) \qquad W^i_{\mathrm{gal}}(z) = b^i_{\mathrm{g}}\, n_i(z) \qquad W(\chi) = W(z)\, \frac{H(z)}{c}$$

Key point

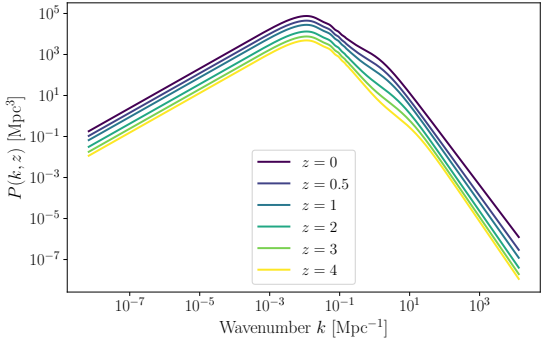
The FRB×galaxy cross-spectrum measures $b_{\mathrm{FRB}}\, b^i_{\mathrm{g}}$ where the two radial kernels overlap ⇒ each tomographic bin probes the FRB bias at a different redshift.

Limber 1953; LoVerde & Afshordi 2008; Khosravi et al. 2016

Pipeline ingredients: Planck 2018 cosmology and nonlinear $P(k, z)$

Parameter	Value
H_0 [km s^{-1} Mpc^{-1}]	67.36
$\Omega_{m,0}$	0.3153
$\Omega_{b,0}$	0.0493
σ_8	0.8111
n_s	0.9649

- Python pipeline: NumPy, SciPy, astropy, `hmf`
- $z \in [0.01, 5]$, 1000 log-spaced points
- $\ell \in [10, 1000]$



Nonlinear matter power spectrum at six redshifts

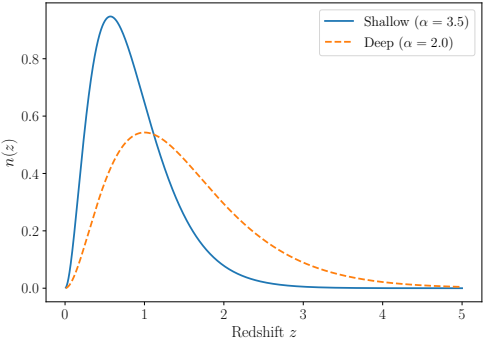
Planck Collaboration 2018; Murray et al. 2013 (`hmf`)

Two FRB survey configurations: shallow and deep

$$n(z) \propto z^2 \exp(-\alpha z)$$

Configuration	α	N_{tot}	f_{sky}
Shallow	3.5	5000	0.9
Deep	2.0	50 000	0.9

- Shallow: CHIME-like
- Deep: SKA-like, **10×** more bursts
- Deep: **≈ 40 %** of bursts at $z > 1.5$



Normalized FRB redshift distributions

Reischke et al. 2021; CHIME/FRB Collaboration 2026; Hashimoto et al. 2020

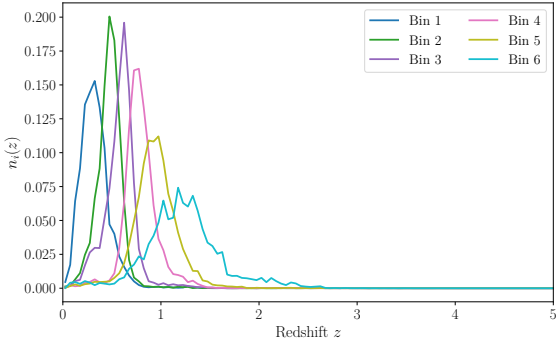
KiDS-Legacy provides six tomographic galaxy bins

- **2.6** m VLT Survey Telescope, Paranal
- **1347** deg^2, i.e. $f_{\mathrm{sky}} = \mathbf{0.0327}$
- Nine bands with VIKING
- r-band limit 24.8 (5σ), seeing **0.7**″
- Photo-z calibrated with **126 000** spectra

Galaxy bias (constant clustering)

$$b_{\mathrm{g}}^{i} = \frac{0.95}{D_{+}(\langle z \rangle_i)}$$

from **1.14** (bin 1, $\langle z \rangle = 0.34$)
to **1.72** (bin 6, $\langle z \rangle = 1.22$)



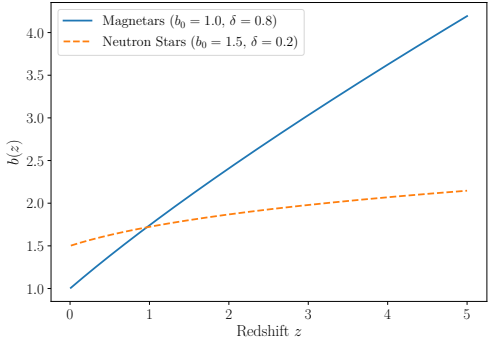
Tomographic redshift distributions of KiDS-Legacy
de Jong et al. 2013; Wright et al. 2024, 2025; Johnston et al. 2025

The progenitor models differ in their bias evolution

$$b(z) = b_0 (1 + z)^{\delta}$$

Progenitor class	b_0	δ
Magnetars	1.0	0.8
Neutron stars	1.5	0.2

- Magnetars: low amplitude, strong evolution (star-forming hosts)
- Neutron stars: high amplitude, weak evolution (massive, old hosts)
- Parameters of interest: $\theta = (b_0, \delta)$

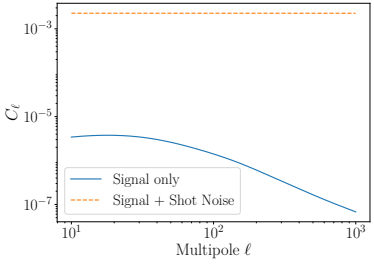


FRB host population bias models

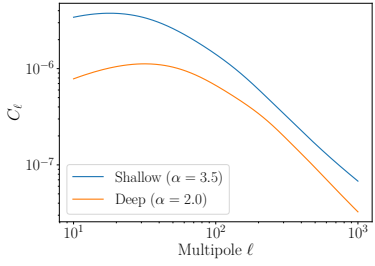
Pan et al. 2020

The FRB auto-correlation alone is shot-noise dominated

$$N_\ell = \frac{1}{\bar{n}} \qquad \bar{n}_{\mathrm{FRB}} = \frac{N_{\mathrm{tot}}}{4\pi f_{\mathrm{sky}}}$$



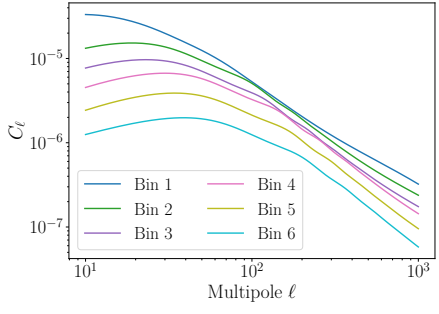
Shallow survey, with and without shot noise



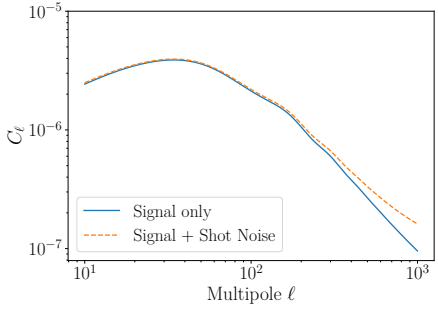
Shallow and deep surveys, signal only

Shallow survey: shot noise exceeds the signal by ~ 3 orders of magnitude at all ℓ

Galaxy auto-spectra: a dense sample with high signal-to-noise



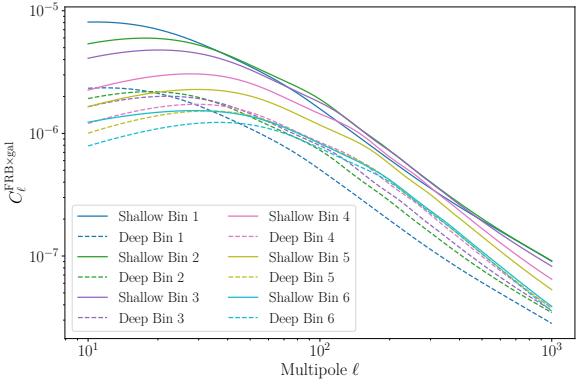
Signal-only spectra, six tomographic bins



Signal and shot noise, bin 5

- Amplitude decreases toward higher bins despite rising bias: broader radial extent, χ^{-2}
- Signal dominates over shot noise except toward the highest multipoles

Cross-correlation: no shot noise and tomographic leverage



FRB×galaxy cross-spectra, magnetar model

- $2 \times 2 \times 6 = 24$ cross-spectra
- Independent samples
  ⇒ no shot-noise term
- Six bins probe the FRB bias at six effective redshifts
- Shallow survey stronger in every bin: low-z weighting of the Limber kernel, broad bin 6

Fisher forecast for the FRB host bias

$$F_{ij} = f_{\mathrm{sky}} \sum_{\ell=\ell_{\mathrm{min}}}^{\ell_{\mathrm{max}}} \frac{2\ell+1}{2} \mathrm{Tr}\left[C_\ell^{-1} \frac{\partial C_\ell}{\partial \theta_i} C_\ell^{-1} \frac{\partial C_\ell}{\partial \theta_j} \right]$$

$$\frac{\partial C_\ell}{\partial \theta_i} = \frac{C_\ell(\theta_i + h_i) - C_\ell(\theta_i - h_i)}{2h_i}$$

$$\mathrm{Cov} = F^{-1} \qquad \sigma_i = \sqrt{\mathrm{Cov}_{ii}}$$

$$\mathrm{FoM} = \frac{1}{\sqrt{\det \mathrm{Cov}}}$$

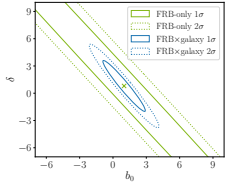
- C_ℓ: matrix of all auto- and cross-spectra, shot noise in the auto-spectra
- $\theta = (b_0, \delta)$, step $h_i = 0.01\,\theta_i$
- Cosmology fixed to Planck 2018

Sky fractions

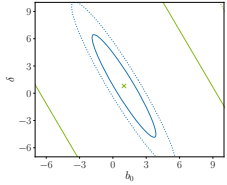
FRB-only: $f_{\mathrm{sky}} = 0.9$ FRB×galaxy: $f_{\mathrm{sky}} = 0.0327$ (KiDS) ⇒ ≈ 28× smaller area

Tegmark et al. 1997; Reischke et al. 2021; Coe 2009; Wang 2008

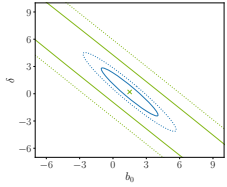
Adding galaxies turns a non-detection into a measurement



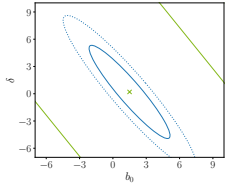
(a) Magnetars, deep



(b) Magnetars, shallow



(c) Neutron stars, deep



(d) Neutron stars, shallow

- Green: FRB auto-correlation only
- Blue: FRB × six KiDS bins
- Solid: 1σ, dotted: 2σ
- Cross: fiducial values

FRB-only contours extend far beyond the plotted range

Constraints improve by more than an order of magnitude

		σ_{b_0}		σ_δ		FoM		
Progenitor	Config.	FRB	FRB×gal	FRB	FRB×gal	FRB	FRB×gal	Gain
Magnetars	Deep	21	1.26	28	1.85	0.032	1.6	51
Magnetars	Shallow	83	1.90	172	3.73	0.001	0.4	377
Neutron stars	Deep	30	1.69	29	1.75	0.020	1.1	60
Neutron stars	Shallow	92	2.41	139	3.39	0.001	0.3	294

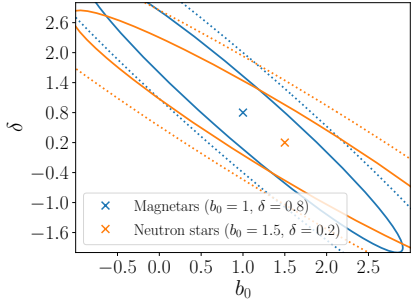
Joint vs. FRB-only

σ reduced by factors 15–46
FoM increased by factors 51–377
despite ≈ 28× smaller sky area

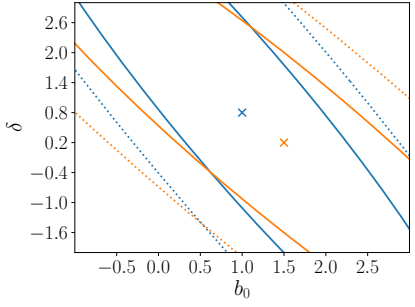
Deep vs. shallow

FRB-only: σ_{b_0} improves 3–4×
Joint: only 1.4–1.5×
≈ 40 % of deep bursts at $z > 1.5$, beyond most KiDS galaxies

KiDS cannot yet distinguish the progenitor channels



(a) Deep survey configuration



(b) Shallow survey configuration

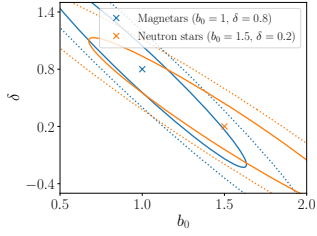
- Fiducial separation $\Delta b_0 = 0.5$, $\Delta\delta = 0.6$ lies well within 1σ
- Relative precision similar: $\sigma_{b_0}/b_0 = 1.3$ (magnetars), 1.1 (neutron stars), deep

Limitations

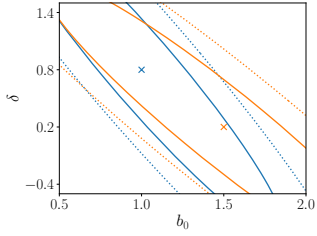
- Limber approximation restricts the analysis to $\ell \geq 10$, excluding the largest scales
- Gaussian likelihood and covariance neglect non-Gaussian contributions
  $\Rightarrow$ uncertainties are optimistic lower bounds
- Power-law bias adequate for $z \lesssim 1.5$, where the constraints come from; at higher z the bias evolves more strongly and depends on halo mass
- FRB redshift distribution and fiducial bias values are model assumptions, not measurements
- Survey geometry only through f_{sky}; a real analysis needs pseudo-spectra

Berti et al. 2021; Pan et al. 2020; Tinker et al. 2010; Hivon et al. 2002

Outlook: LSST Y10 brings discrimination within reach



(a) Deep



(b) Shallow

- 18 000 deg^2, f_{sky} = 0.4363, ≈ 13× KiDS
- Ten-bin Y10 lens sample
- FoM × 9–12
- σ_{b_0}: 1.26 → 0.42 (deep, magnetars)

- Deep: neutron-star fiducial just outside the magnetar 1σ contour; both within 2σ
- Shallow: models still overlap at 1σ
- FoM gain below 13: LSST lens bins end at $z \approx 1.2$, KiDS bin 6 reaches beyond $z = 1.5$

Ivezić et al. 2019; Zhang et al. 2025

Conclusion

1. Forecasting pipeline: nonlinear $P(k, z) \rightarrow$ Limber auto- and cross-spectra of FRBs and six KiDS-Legacy bins $\rightarrow$ Fisher forecast for b_0 and δ
2. FRB auto-correlation alone is shot-noise dominated and does not constrain the bias
3. Cross-correlation with KiDS: σ reduced 15–46×, FoM increased 51–377×, despite a $\approx$ 28× smaller sky area
4. KiDS footprint not yet sufficient to separate prompt magnetars from delayed neutron stars
5. LSST Y10: FoM × 9–12, separation becomes comparable to the uncertainties

Take-home message

FRB×galaxy cross-correlation is a promising probe of FRB progenitors, and next-generation galaxy surveys can make it decisive.

Thank you for your attention!

Backup: fiducial cosmology and KiDS tomographic bins

Parameter	Symbol	Value
Hubble constant	H_0 [km s^{-1} Mpc^{-1}]	67.36
Total matter density	$\Omega_{m,0}$	0.3153
Baryon density	$\Omega_{b,0}$	0.0493
Fluctuation amplitude	σ_8	0.8111
Scalar spectral index	n_s	0.9649
CMB temperature	$T_{CMB,0}$ [K]	2.7255

Planck 2018 TT,TE,EE+lowE+lensing

Bin	$\langle z \rangle_i$	b_g^i
1	0.335	1.135
2	0.477	1.221
3	0.587	1.291
4	0.789	1.423
5	0.940	1.525
6	1.224	1.721

$b_g^i = 0.95/D_+(\langle z \rangle_i)$

Backup: full-sky spectrum and the Limber approximation

Exact full-sky angular cross-power spectrum

$$C_\ell^{AB} = \frac{2}{\pi}\int_0^\infty k^2\,\mathrm{d}k \int \mathrm{d}\chi\, W^A(\chi)\, j_\ell(k\chi) \int \mathrm{d}\chi'\, W^B(\chi')\, j_\ell(k\chi')\, P_{AB}(k;\chi,\chi')$$

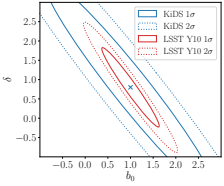
- Spherical Bessel functions j_ℓ peak sharply at $k\chi \approx \ell + 1/2$ for large ℓ
- Kernels vary slowly compared to j_ℓ $\Rightarrow$ only $k = (\ell + 1/2)/\chi$ contributes

$$\Rightarrow \quad C_\ell^{AB} \approx \int \frac{\mathrm{d}\chi}{\chi^2}\, W^A(\chi)\, W^B(\chi)\, P_{AB}\left(\frac{\ell + 1/2}{\chi},\, z(\chi)\right)$$

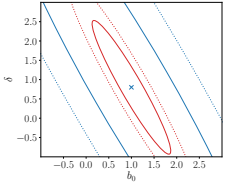
- Valid for $\ell \geq 10 \Rightarrow \ell_{\mathrm{min}} = 10$
- $\ell_{\mathrm{max}} = 1000$: nonlinear $P(k)$ prescription still sufficiently reliable

Limber 1953; LoVerde & Afshordi 2008

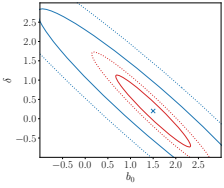
Backup: KiDS versus LSST Y10



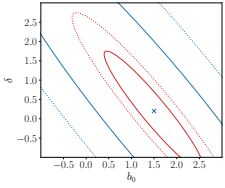
(a) Magnetars, deep



(b) Magnetars, shallow



(c) Neutron stars, deep

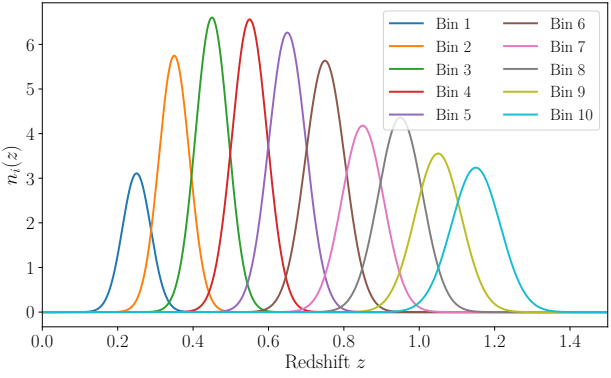


(d) Neutron stars, shallow

	FoM		
	KiDS	LSST	Ratio
Mag. deep	1.6	14.8	9
Mag. shallow	0.4	4.9	12
NS deep	1.1	11.5	10
NS shallow	0.3	4.2	12

LSST	σ_{b_0}	σ_δ
Mag. deep	0.42	0.68
Mag. shallow	0.57	1.14
NS deep	0.55	0.61
NS shallow	0.73	1.02

Backup: LSST Y10 lens sample



Tomographic redshift distributions of the LSST Y10 lens sample (ten bins)

Zhang et al. 2025

Backup: an FRB compared to a mobile phone on the Moon

GSM-900 phone: $P = 2\,\mathrm{W}$, channel width $B = 200\,\mathrm{kHz}$, isotropic emission at $d \approx 3.84 \times 10^8\,\mathrm{m}$

$$F_{\mathrm{phone}} = \frac{P}{4\pi d^2} \approx 1.08 \times 10^{-18}\,\mathrm{W\,m^{-2}} \qquad S_{\mathrm{phone}} = \frac{F_{\mathrm{phone}}}{B} \approx 540\,\mathrm{Jy}$$

- Typical FRB: $S_{\mathrm{FRB}} \approx 0.54\,\mathrm{Jy}$
- $S_{\mathrm{phone}}/S_{\mathrm{FRB}} \approx 10^3$
- Order-of-magnitude illustration; other standards and fluxes give different ratios

ETSI/3GPP TS 45.005; Petroff et al. 2019; Billings 2013

Backup: further extensions

- Replace the parametric $n(z)$ by the measured distribution of the second CHIME/FRB catalog, already of shallow-configuration size
- Full likelihood sampling instead of the Fisher formalism: non-Gaussian posteriors and covariance contributions
- Scale-dependent halo-model bias linked to host halo masses, extending to smaller scales
- Realistic survey masks: pseudo-C_ℓ estimation with mode-coupling matrices

CHIME/FRB Collaboration 2026; Hivon et al. 2002